

ADITYA DNYANESHWAR DAHALE

Systems & Backend Software Engineer (2+ Years Experience) | Founder @ Spryzen | Rust, Go, Python, Linux Internals

Location: Pune, Maharashtra, India • Notice Period: **Immediate Joiner (0 Days Notice)** • Phone: +91 9021574315 • Email: adityadahale96@gmail.com
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PROFESSIONAL SUMMARY

High-performance Systems & Backend Software Engineer with **2+ years of production experience** in low-latency distributed runtimes, transparent network proxying, and kernel bypass architectures. Founder of **Spryzen** (Govt. registered Udyam startup), architecting zero-copy inspection engines delivering 6.92M PPS and sub-microsecond latency. Production open-source contributor to Bytecode Alliance Wasmtime, Tokio, Cloudflare Pingora, smoltcp, Foundry, and Hyper. Based in Pune, available to join immediately (0 days notice).

PROFESSIONAL WORK EXPERIENCE

- **Founder & Lead Systems Engineer — Spryzen (Govt. Registered Udyam MSME Startup)** 2024 – Present (2+ Yrs) | Pune, India
 - **Spryzen+ Network Telemetry & Proxy:** Architected kernel-bypass eBPF/XDP proxy pipeline delivering **6.92M PPS at 144ns latency** (4.83M sustained RPS, 0.207µs P99); engineered LPM Trie CIDR blocking, token-bucket rate limiter, and zero-allocation JA4 TLS fingerprinting parser.
 - **IronWall+ WAF Engine:** Built zero-allocation security inspection engine achieving **4.83M RPS with 27.99ns fast-path normalization** using 256-byte O(1) RISKY_CHAR LUT and SIMD acceleration; verified under multi-core K6 load testing with CPU/memory flame graphs.
 - **Iron-Proxy:** Engineered high-throughput distributed RPC routing proxy in asynchronous Rust (Tokio) with tiered LRU memory caching and native Prometheus observability exports.
- **Core Systems & Infrastructure Contributor — Open Source Runtimes** 2024 – Present (2+ Yrs) | Global
 - Authored and merged mission-critical fixes across production runtimes powering AWS, Cloudflare, Fastly, and Paradigm. Resolved kernel socket busy loops, memory leaks, and protocol violations passing strict multi-threaded Loom and regression test suites.

VERIFIED OPEN-SOURCE CONTRIBUTIONS (CLICK TO VIEW REPOSITORY & PR)

- **bytecodealliance / wasmtime** | [PR #14387](#) — **POSIX Trailing Slash Symlink Resolution** ✓ MERGED
Fixed directory resolution in WASI filesystem engine when FollowSymlinks::No is set and trailing slash requires directory resolution, preventing erroneous ENOTDIR errors; merged by Alex Crichton.
- **tokio-rs / tokio** | [PR #8405](#) — **Semaphore Integer Overflow & Permit Leak** ✓ MERGED
Resolved critical permit leak in SemaphorePermit::merge where counts exceeding u32::MAX wrapped; migrated to checked usize, passing Tokio's Loom CI.
- **foundry-rs / foundry** | [PR #16707](#) — **Chisel NTFS Stream Isolation & Traversal Guard** ✓ MERGED
Hardened session cache in crates/chisel against Windows NTFS Alternate Data Streams and path traversals, preventing arbitrary disk write escapes.
- **superfly / corrosion** | [PR #559](#) — **PostgreSQL Wire AST Rollback Handling** ✓ MERGED
Implemented PostgreSQL wire protocol transaction abort state machine (SQLSTATE 25P02) and Extended Query sync recovery in corro-pg distributed SQLite.
- **smoltcp-rs / smoltcp** | [PR #1194 & #1208](#) — **Zero-Alloc Wire Parser & Padding Fix** ✓ MERGED • CI Green
Engineered single-pass zero-allocation TcpOptionsSummary parser ([#1194](#)); resolved datagram can_recv() false positives by draining padding to eliminate 100% CPU busy loops ([#1208](#)).
- **FoxIO-LLC / ja4** | [PR #312 & #314](#) — **Zero-Allocation Header Filter & Raw Hex Digest** ✓ 2x MERGED
Eliminated heap allocations during HTTP/1 header filtering ([#312](#)); direct hex-encoded 6-byte raw digest slices in ja4/ja4x, removing 2 allocations and avoiding encoding 26 unused bytes ([#314](#)).
- **oschwald / maxminddb-rust** | [PR #120](#) — **Native Big-Endian Byte Decoding** ✓ MERGED
Accelerated GeoIP binary lookups using native big-endian byte decoding (u32::from_be_bytes), enabling single-CPU-instruction decoding of 32-bit database nodes.
- **cloudflare / pingora** | [PR #1000 & #996](#) — **Parser Admission Limits & RFC 9110 Timeout** Reviewed • CI Green
Engineered configurable HTTP/1 parser limits ([#1000](#)); mapped upstream connect, TLS handshake, and read/write timeouts to 504 Gateway Timeout ([#996](#)).
- **hyperium / hyper** | [PR #4183](#) — **RFC 6585 max_header_size Protocol Enforcement** In Review • Sean McArthur
Added server/client builder APIs, strict RFC 6585 §5 (431 Too Large) rejection, stream buffer zero-copy guards, and chunked trailer decoder limits across 180+ tests.
- **tursodatabase / libsql** | [PR #2280](#) — **Default WAL Journal Concurrency** CI 100% Green
Enforced Write-Ahead Logging (WAL) concurrency as default for local SQLite instances, preventing database locks under concurrent access across 12 CI test suites.

TECHNICAL SKILLS & DOMAIN EXPERTISE

- **Systems & Languages:** Rust (Tokio, Loom, Atomics, Lock-Free, Zero-Copy Parsing), Go, Python (FastAPI, Django), C/C++, Bash, SQL
- **Kernel & Networking:** eBPF/XDP, Linux Networking Internals, TCP/IP Stack, Sockets, HTTP/1.1–HTTP/3, TLS/JA4 Fingerprinting, Reverse Proxies (Pingora, Nginx, Envoy)
- **Storage & Databases:** PostgreSQL, Redis, SQLite/libSQL (WAL Journal Concurrency), Distributed Key-Value Caching, Vector Storage
- **Cloud & Reliability:** AWS (EC2, S3, CloudWatch), Docker, Linux System Administration, GitOps, CI/CD (GitHub Actions), Prometheus, Grafana, Fuzz Testing

VERIFIED INDUSTRY CERTIFICATIONS

- **AWS Cloud Practitioner Essentials** — Amazon Web Services | **AWS AI Fundamentals & Cloud** — AWS / Coursera | **Developing AI Apps with Python & Flask** — IBM
- **Python for Data Science, AI & Development** — IBM | **Microsoft Python Programming Fundamentals** — Microsoft | **Generative AI: Evolution of Online Search** — LinkedIn

EDUCATION

- **Savitribai Phule Pune University (SPPU) — Bachelor of Engineering (B.E.) in Computer Engineering** Pune, India
Cumulative SGPA: 8.32 / 10.0 (1st Sem: 8.27 | 2nd Sem: 8.36) | Core Focus: Systems Programming, Operating Systems, Computer Architecture, Data Structures